

Fabio Canavarro

Systems Programmer & Developer

Hsinchu, Taiwan | fabioc.me | github.com/FabioCanavarro | [LinkedIn](#)

EDUCATION

National Tsing Hua University (NTHU)

International Bachelor Degree Program (IBP) — EECS

Hsinchu, Taiwan

Expected 2030

Prime One School

High School Diploma — 93.27% Average

Medan, Indonesia

Graduated 2026

OPEN SOURCE CONTRIBUTIONS

NTHU Mods | *React, Bun, Docker*

- Contributed to the university's open-source platform by developing and shipping a real-time library seat vacancy tracker utilizing public APIs, alongside a laundry availability feature.

Systems Ecosystem | *eBPF, Kubernetes, Infrastructure*

- Submitted pull requests to **rwat** (eBPF threat detection), **Scaligator** (intelligent Kubernetes HPA alternative), and **infraust** (high-efficiency Minecraft server host).

PROJECTS

EpochDB | *Rust, Sled, Prometheus*

- Built a concurrent key-value database over the Sled storage engine; published to crates.io.
- Implemented atomic TTL expirations and integrated Prometheus for observability metrics.

Irridation | *Rust, Language Virtual Machine, Assembly*

- Engineered a custom assembly language featuring a two-phase assembler and a bespoke register-based virtual machine (VM) built in Rust.
- Implemented an interactive REPL and CLI for live, low-level bytecode execution.

UnitOS | *Bare-Metal Rust, x86 Architecture, QEMU*

- Developed a custom operating system kernel entirely in bare-metal Rust.
- Configured the GDT, handled hardware interrupts, and built custom paging for memory management.

FerrisLog | *Rust, Networking, Distributed Systems*

- Engineered a log-structured persistent key-value store entirely from scratch in Rust, featuring state snapshots and a custom client-server network protocol.

Chrono Domain | *Java, Minecraft Fabric*

- Developed a Minecraft mod to manipulate the flow of time for entities and blocks in different chunks, creating a visual time-dilation zones; attained hundreds of downloads on Modrinth.

NeuraLearn | *C++, ESP32-S3, FreeRTOS*

- Prototyped an AI smart desk leveraging dual-core task scheduling to synchronize hardware modules.

AgroBioSync | *C++, ESP8266, React*

- Automated an IoT agricultural control system; awarded 1st place in a high school research competition.

Mochadeck & Hal-Luminance | *KiCad, Bare-Metal Rust, Onshape*

- Designed a custom mechanical keyboard PCB schematic and wrote a bare-metal PWM controller for 12V COB lighting.

TECHNICAL SKILLS

Languages: Rust, C++, Java, Python, TypeScript, JavaScript, Assembly

Hardware & Embedded: ESP32, ESP32-S3, ESP8266, Arduino, KiCad, PlatformIO

Developer Tools: Arch Linux, Neovim, Vim, VS Code, PyCharm, Git, Docker, QEMU, Vercel

Spoken Languages: English (IELTS 8.0), Indonesian, Mandarin (TOCFL A2, HSK 3)